

INTERNSHIP REPORT

Submitted by

Malempati Teja
Supraja Technologies employee id: ST#IS#7637

Under the Supervision of

UPENDRA
Senior Security Analyst

Krishna
Security
Analyst



Registered And Head Office D.NO: 11-
9-18, 1st Floor,
Majjivari Street, Kothapeta,
Vijayawada - 520001.

+91 9550055338 / +91 7901336873

contact@suprajatechnologies.com

COMPANY INTRODUCTION:

Supraja Technologies is a leading Knowledge and Technical Solutions Provider and pioneer leader in IT industry, is operating based out of Vijayawada.

R&D at Supraja

With a 24X7 work in Research & Development, experts at Supraja Technologies work under our :

- **Supraja Technologies Cyber Security Cell**

About Supraja Technologies:

Supraja Technologies (a unit of CHSMRLSS Technologies Pvt. Ltd.) with its foundation pillars as Innovation, Information and Intelligence is exploring indefinitely as a **Technology Service Provider (Corporate Consulting)** and as a **Training Organization (Ed-Tech)** as well.

You may visit us at :

www.suprajatechnologies.com

The multi domains of trainings which Supraja Technologies operate include the following :

- **Workshops & Hackathons**
 - Engineering Colleges
 - Corporate Companies (Startups & MNC's)
 - Government Organizations

- **Classroom Trainings Cum Certification Courses**

- Summer Training (30-45 Days)
- Winter Training (10 - 15 Days)
- Weekend Training (2 Days)
- 1 Month / 3 Months / 6 Months Courses

- **On-site Trainings**

- Value Added Courses / Two Credit Courses for Colleges
- Faculty Development Programs
- College Summer Training (15 Days, 30 Days, 45 Days & 60 Days)
- Govt Agencies, Police Academies, Corporates etc

- **Cloud Campus**

- (Distance Learning Program) *Coming Soon

- **Internships**

- Internship for Engineering Students (30 Days, 45 Days & 60 Days)
- Internship for Graduates (6 Months)

- **Lab Setup**

- Cyber Lab

- **CoE**

- Cyber Security Centre of Excellence

- **Supraja Technologies Security Assessment Product**

- (Our SaS Product is currently under development) *Coming Soon

Why Supraja Technologies:

Be it Training or a workshop, the course content is always from R&D Cell of Supraja.

- A proven track record of delivering quality services.
- **68,500+** Students trained by our trainers till date.
- Training Partners of recognized institutions.
- Trainers with excellent research and teaching pedagogy illustrate their findings through corporate standard **practical demonstrations** during their sessions.
- Easy to learn and **hands-on sessions** are given, with additional benefits of Study Material, Tool kit and immediate query handling.
- Self-Prepared **Cyber Security Cell**.
- Supraja Technologies has the best, experienced and highly **skilled bunch of R&D Engineers, Security Analysts, Security Consultants & Trainers**.
- We provide training in Innovating and Trending Technologies to Govt. Officials, Corporate Houses and Colleges.

✓ Something we are proud of :

1. Supraja Technologies CEO Mr.Santosh Chaluvadi is an Alumni of Potti Sriramulu Chalavadi Mallikharjuna Rao College of Engineering and Technology, Vijayawada.

With our CEO this college conducted/organised a 50 hours Nonstop Marathon Training Workshop on Ethical Hacking & Cyber Security for which this respective college and our CEO both holds their name in **“LIMCA BOOK OF RECORDS 2017”**

2. We are very happy to inform you all that our company, Supraja Technologies has been shortlisted for **"Top 50 Tech Companies" award 2019**, conferred at InterCon - Dubai, UAE.

Supraja Technologies is one out of thousands companies that were initially screened by InterCon team of 45+ research analysts over a period of three months and the final shortlist includes 150+ firms and we are very proud to

inform you all that our company Supraja Technologies also happens to be a part of the same.

✓ **Life changing solution/service :**

After working on R&D for around 2 years, finally in the mid 2019 we have successfully developed a service/solution of various techniques and strategies for the Film Industry through which he can kill piracy of any film in online up to 35% right now. This betaservice is being appreciated & adopted by various Tollywood Film Industry Producers & Hero's to safeguard their film from piracy in online and to gain more profits.

By the end of 2033 our vision is to rollout a complete full packed service/solution where we can kill piracy entirely 100% everywhere in online for sure.

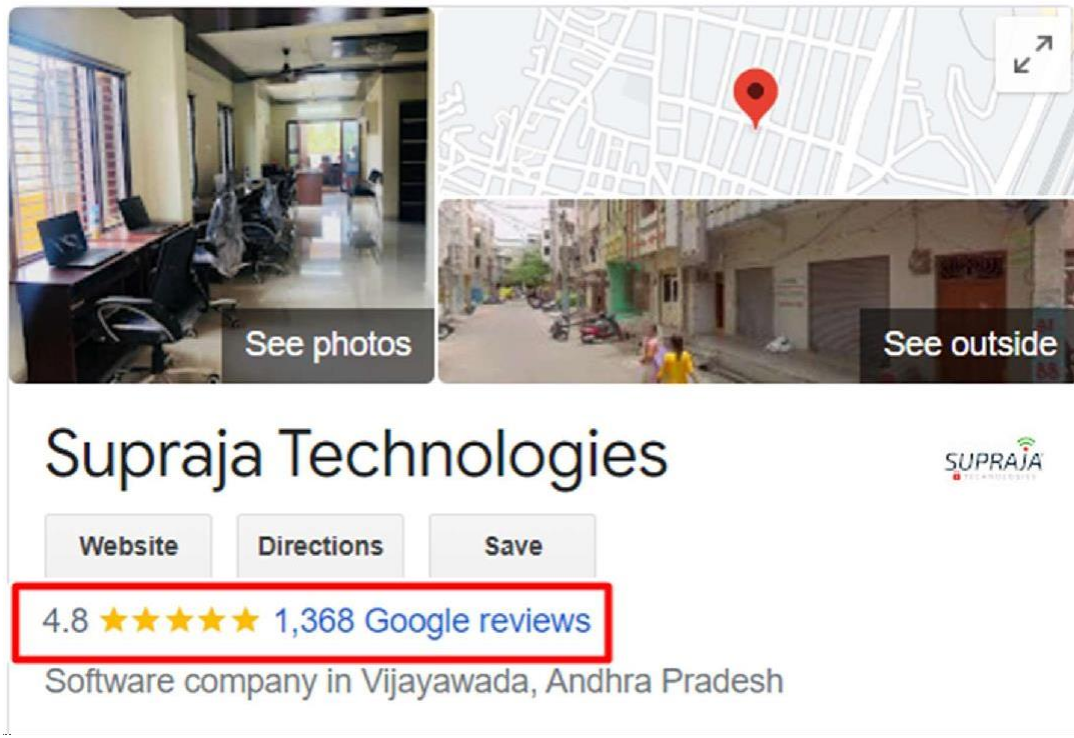
Appreciation :

Received a great appreciation from our 1st Tollywood Film Industry client Mr.Saptagiri for providing our Anti-Piracy betaservice for his film VAJRA KAVACHADARA GOVINDA

✓ **Achievements:**

- On 17th August 2024, We (Supraja Technologies) launched our company's **Centre of Excellence in Cyber Security (CoE)** at Ramco Institute of Technology, Rajapalayam
- On 18th September 2024, We (Supraja Technologies) launched our company's **Centre of Excellence in Cyber Security (CoE)** at SRM University (Ramapuram Campus), Chennai
- On 20th November 2024, We (Supraja Technologies) launched our company's **Centre of Excellence in Cyber Security (CoE)** at St.Joseph's Institute of Technology, Chennai

**Our Company Supraja Technologies
is one of the emerging startup
in Andhra Pradesh
with **4.8** Google Ratings**



Link : <https://bit.ly/SuprajaGoogle>

AND

**also check our Company CEO Instagram Profile
for our recent more success stories:**

<https://www.instagram.com/chaluvadisantosh/>



Santosh Chaluvadi

**Founder & CEO
Supraja Technologies**

He is a 28-year-old entrepreneur, one of the India's efficient Cyber Security Analyst and also he is an expert Digital Marketer as well. He is a digital marketer by profession and security enthusiast by passion. He primarily focuses on content building, testing and monetization of blogs. He has successfully developed many websites and done the security testing himself to ensure that the user's data is in safe hands and their privacy is protected. He is very active on social media and shares lot of tech stuff with his followers. The young student hacker has solved many issues with the vulnerabilities present in various websites and databases, given a solution in clearing the loopholes in order to protect the data to be leaked from the databases. Besides Ethical Hacking & Cyber Security, he also has a passion in Blogging & Digital Marketing.

While pursuing his engineering itself, he has trained many young generation people/students of more than 3500+ from various parts across Andhra Pradesh through his workshops, seminars, courses in Cyber Security and this makes him one of the youngest student trainers in India.

At the age of 20 he conducted his first workshop on Blogging & Ethical Hacking which was the beginning to his success in this field and right now he has handful of workshops to train students, government and corporate organizations as well in Andhra Pradesh & Telangana. He is the only student trainer who started conducting workshop for his peers, professors and for corporates.

The year 2016 gave me the conviction and confidence to notch up whatever I was doing. I'd organized a 50-hour marathon event on Ethical Hacking and Cyber Security in PSCMR College, Vijayawada that went on to bag to achieve Limca Book of Records for non-stop longest duration workshop. The impact we were making was clearly visible by now. Diverse people from Jammu & Kashmir in the North to Kanyakumari in the South had attended the event. Some of the Government of India officials took notice of this record and invited our team for couple of conferences at New Delhi. The country's esteemed institutions were reaching out to me to conduct training, workshops, and events on a myriad of subjects related to online security. Multi-National Corporations (MNC's) had begun to consult me on the Cyber Security of their systems.

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Centre of Excellence in Cyber Security (CoE) at St.Joseph's Institute of Technology, Chennai

❖ Records, Appreciations, Awards & Recognitions etc at a glance :

1. Holds a National Record in "Limca Book of Records – 2017"
2. Ex-Associate Member for National Cyber Safety and Security Standards (NCSSS)
3. Steering Committee Member for United Conference on Cyber Space (UNITEDCON 2020)
4. Judge for the Grand Finale of SIH (Smart India Hackathon 2024) Software Edition for the Nodal Centre at Sri Sai Ram Engineering College, Chennai which is an initiative by Ministry of Education (Govt. of India) and AICTE
5. Received Appreciation from Mr.Amit Narayan, Executive Director at Rajahmundry Asset of Oil and Natural Gas Corporation Limited (ONGC) on 16th December, 2022 for training their employees on Cyber Security
6. Awarded as a "Karmaveer Chakra - 2019", on 12th October 2019 at IIT Delhi, which was instituted by iCONGO in partnership with the United Nations
7. Received Appreciation from Mr.Sandeep Rathore, Commissioner of Police, Chennai on 2nd March 2024 for the exceptional commitment and invaluable contribution as a Member of JURY at Greater Chennai Police Cyber Hackathon 3.0
8. Evaluator for the "Innovative Bharat" which is organized by AICTE and Ministry of Education held on 6th January 2024
9. Judge for the Grand Finale of SIH (Smart India Hackathon 2023) Senior Software Edition for the Nodal Centre at PVPSIT, Vijayawada which is an initiative by Ministry of Education (Govt. of India) and AICTE
10. Appointed as a Member for Board of Studies on 19th April 2025 for the Departments of Cyber Security, Data Science and AIML at Bapatla Engineering College, Bapatla
11. Appointed as a Member for Board of Studies on 12th March 2025 for the Department of Cyber Security at St. Joseph's Institute of Technology, Chennai
12. Appointed as one of the Industry Academia Advisory Council Member on 30th September 2023 for the Department of Cyber Security at VNR Vignana Jyothi Institute of Engineering & Technology, Hyderabad
13. Appointed as a Member for Board of Studies on 4th May 2023 for the Department of Cyber Security at Madanapalle Institute of Technology & Sciences, Madanapalle
14. Appointed as a Member for Board of Studies for the Department of MCA at NRI Institute of Technology, Perecherla (Guntur)
15. Awarded as a "Social Media Influencer - 2019", on 30th June 2019 by Jignasa in association with Government of Andhra Pradesh
16. Nominated for "INDIA 500 CEO AWARD 2019"
17. Invited & Interviewed by ETV Andhra Pradesh news channel on 27th July, 2019 for a Special Story Interview on "Spy Apps"

18. Appreciated by Mr.Sridhar, Sub-Inspector of Police at Central Crime Branch, Vijayawada on 23rd October, 2018 for exclusively training him on Special Investigation Course, which will help him to solve the cyber crime cases easily
19. Received a great appreciation from our 1st Tollywood Film Industry client Mr.Saptagiri, for providing Anti-Piracy betaservice for his movie VAJRA KAVACHADARA GOVINDA

✓ **Something we are proud of :**

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Some Glimpses of our Journey



Mr.Santosh Chaluvadi – CEO, Supraja Technologies
Giving hands-on Cyber Security training workshop to the CSE students
at IIT Kharagpur



Mr.Santosh Chaluvadi – CEO, Supraja Technologies was
Invited & Interviewed by ETV Andhra Pradesh news channel on 27th July, 2019
for a Special Story Interview on “Spy Apps”



Mr.Santosh Chaluvadi – CEO, Supraja Technologies was awarded as a **"Social Media Influencer 2019"** in recognition of his remarkable achievements in the social media as a part of First International Social Media Festival on 30th June 2019 by Jignasa in association with Government of Andhra Pradesh



On 23rd October 2018 Mr.Santosh Chaluvadi, CEO - Supraja Technologies and Mr.Krishna Chaitanya, CTO - Supraja Technologies had successfully completed delivering Special Investigation Course training in Cyber Security to Mr.Sridhar Garu, Sub-Inspector of Police at Central Crime Branch, Vijayawada which will help him to solve the cyber crime cases easily



ETV Andhra Pradesh News Channel interviewed Mr.Santosh Chaluvadi, CEO - Supraja Technologies for his achievements in the domain of Cyber Security & Digital Marketing



Supraja Technologies was invited by Indian Air Force (Air Wing NCC) to deliver a session on Latest Cyber Crimes & Awareness for the NCC cadets, staff and officers on 4th July, 2019



Supraja Technologies – CEO, CTO & CMO with
Indian Air Force (Air Wing NCC) Group Captain Sandeep Gupta.
We thank Mr.Sandeep Gupta for inviting us on 4th July, 2019 to deliver a session on Latest
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Got it — here's your text with only the sentence structure changed w keeping the exact meaning intact:

1. Phases of Hacking

Hacking is usually performed through a set of organized stages. Understanding these steps allows security experts to predict, identify, and stop attacks more effectively.

A. Reconnaissance (Information Gathering):

The process begins with reconnaissance, which is often the most important stage. At this point, the attacker gathers as much information as possible about the target before making any intrusion attempt. This information can include IP addresses, domain information, network layout, system details, or employee data. Reconnaissance can be done in two ways:

- **Passive:** Information is collected without directly contacting the target (for example, through WHOIS lookups or by examining social media profiles).
- **Active:** The attacker directly interacts with the target's systems to gather data (such as performing ping sweeps or DNS queries).

B. Scanning:

Once basic details are obtained, the attacker scans the target to locate live systems, detect open ports, identify running services, and find possible vulnerabilities. Tools like Nmap, Nessus, or OpenVAS are used here to map out the attack surface. The main purpose is to discover potential points of entry.

C. Gaining Access:

At this stage, the attacker exploits the identified weaknesses to break into the system. Methods may include password cracking, exploiting software vulnerabilities, or injecting malicious code into applications. The goal is to gain either user or administrative control.

D. Maintaining Access:

After gaining entry, the attacker ensures they can return to the compromised system in the future. This is done by installing backdoors, rootkits, or Remote Access Trojans (RATs), which allow access even if the original vulnerability is patched. This step is essential for ongoing surveillance or long-term data theft.

A. Covering Tracks:

In the final stage, the hacker attempts to erase any evidence of their presence. This may include deleting logs, hiding malicious files, and using encryption or anonymization techniques to avoid detection by security teams.

2. Phases of Ethical Hacking:



A. Reconnaissance

Also called the information gathering phase, this is where the ethical hacker collects as much data as possible about the target. The goal is to identify potential entry points.

- **Passive Recon:** Gathering publicly available information without direct interaction (e.g., WHOIS, Google Dorking, social media).
- **Active Recon:** Direct engagement with the target systems (e.g., ping sweeps, DNS queries).

B. Scanning

In this phase, the ethical hacker uses technical tools to discover live hosts, open ports, running services, and

vulnerabilities. This step maps the attack surface for potential exploitation.

- **Tools:** Nmap, Nessus, OpenVAS
- Scans include port scanning, vulnerability scanning, and service enumeration.

C. Gaining Access:

This is where the hacker attempts to exploit discovered vulnerabilities to enter the system.

- Methods include password cracking, SQL injection, exploiting software flaws, or phishing simulations.
- The aim is to show how an attacker might breach the system's defences.

D. Maintaining Access

After gaining entry, the ethical hacker tests whether long-term access can be maintained. This simulates persistent threats that remain in the system unnoticed.

- Techniques: Installing backdoors, creating hidden accounts, or simulating malware persistence.

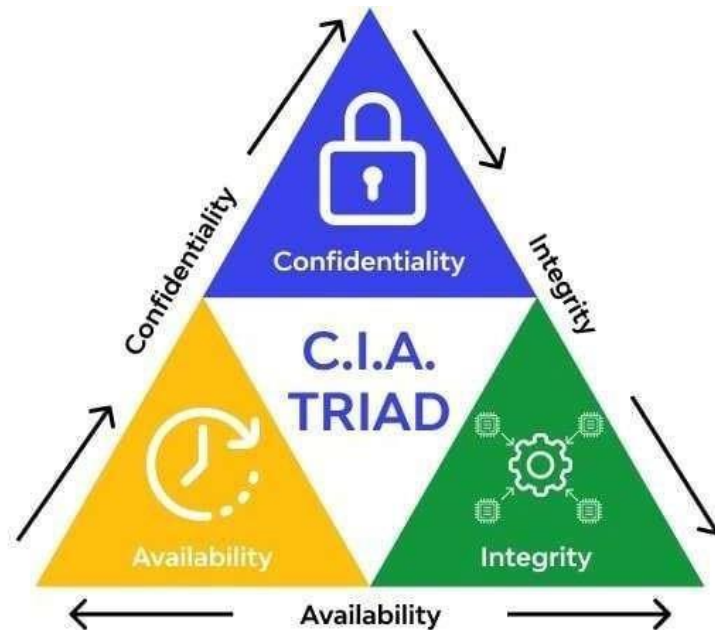
E. Covering Tracks:

The hacker simulates how a malicious actor might erase evidence of their activity.

- This includes clearing logs, deleting temporary files, and hiding malicious scripts.
- In ethical hacking, this is only demonstrated, not fully executed, to avoid damaging the system.

3. Elements of Information Security (CIA Triad):

The CIA Triad—Confidentiality, Integrity, and Availability—is the foundational model for designing and evaluating information security policies. It ensures that information systems remain protected against unauthorized access, alterations, and disruptions.



A. Confidentiality:

Confidentiality ensures that sensitive information is accessible only to authorized individuals or systems. It prevents unauthorized disclosure that could harm an organization or individual.

Implementation Measures:

- Data encryption (AES, RSA)
- Multi-factor authentication (MFA)
- Role-based access control (RBAC)
- Access control lists (ACLs)

Example: Only the finance department has access to company payroll data.

B. Integrity:

Integrity ensures the accuracy, consistency, and reliability of data throughout its lifecycle. It prevents unauthorized modification or deletion of information, whether intentional or accidental.

Implementation Measures:

- Cryptographic hashing (SHA-256, MD5 for checksums)
- Digital signatures and certificates
- Database constraints and version control

Example: Preventing tampering with transaction amounts in online banking systems.

c. Availability:

Availability ensures that authorized users have timely and reliable access to information and resources. It focuses on minimizing downtime and preventing service interruptions.

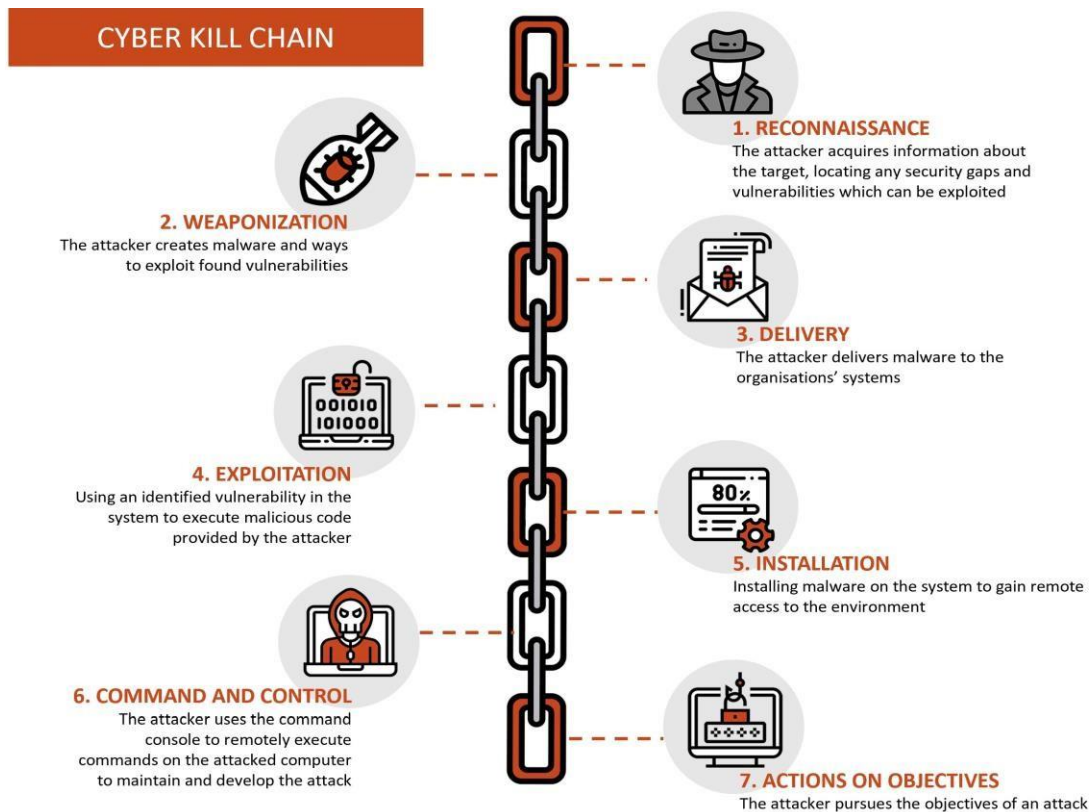
Implementation Measures:

- Load balancing and redundancy
- Regular backups and disaster recovery plans
- Protection against DoS/DDoS attacks

Example: An e-commerce platform remains operational during high-traffic sales events.

4. Cyber Kill Chain Framework:

The Cyber Kill Chain, developed by Lockheed Martin, is a structured model used to understand and counter cyberattacks. It breaks down an attack into sequential stages, helping security teams detect and disrupt threats at each step.



A. Reconnaissance:

The attacker gathers information about the target, such as IP ranges, network structure, technologies used, and employee details.

Example: Using WHOIS, social media, or scanning tools to identify potential entry points.

B. Weaponization:

The attacker creates a malicious payload tailored for the target. This often combines malware with an exploit.

Example: Crafting a malicious PDF that contains an exploit for a known software vulnerability.

C. Delivery:

The attacker sends the malicious payload to the target.

Example: Phishing emails, infected USB drives, or malicious websites.

D. Exploitation:

The payload is triggered, exploiting a vulnerability to execute malicious code on the target system.

Example: Exploiting an unpatched application to run unauthorized commands.

E. Installation:

The attacker installs backdoors, trojans, or other persistence mechanisms to maintain access.

Example: Installing a remote access trojan (RAT) for long-term control.

F. Command and Control (C2):

The compromised system connects to the attacker's command server to receive instructions.

Example: Using encrypted communication channels to send stolen data or receive commands.

G. Actions on Objectives:

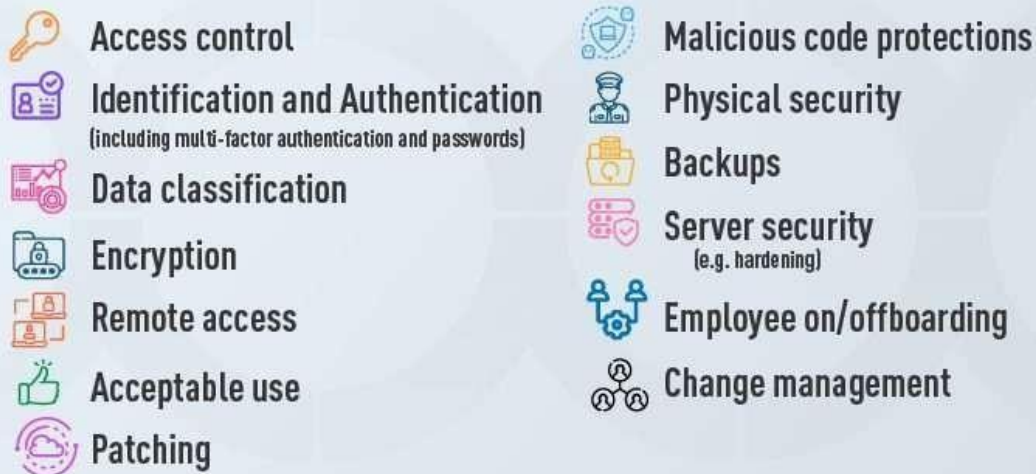
The attacker carries out their final goal, such as data theft, system destruction, or espionage.

Example: Exfiltrating customer data from a company's database.

5. Security Policies in MNC's and MANG:

Security policies in Multinational Corporations (MNCs) and Major American Networking Giants (MANG – Microsoft, Apple, Netflix, Google) are formal guidelines designed to protect digital assets, ensure compliance with regulations, and maintain operational continuity across global branches. These policies cover data protection, access management, incident response, and regulatory compliance.

INFORMATION SECURITY POLICY INCLUSIONS



A. Access Control Policies:

Define who can access specific systems, data, and resources, based on job roles and responsibilities.

- **Implementation:** Role-Based Access Control (RBAC), Multi-Factor Authentication (MFA), and least privilege principle.
- **Example:** Google enforces hardware security keys for all employees.

B. Data Protection Policies

Ensure the confidentiality, integrity, and availability of sensitive data, whether stored, in transit, or in use.

- **Implementation:** Encryption (AES-256), data masking, and secure backups.
- **Example:** Microsoft encrypts all customer data by default in Azure cloud services.

C. Acceptable Use Policies (AUP):

Define how employees can use company devices, networks, and internet resources.

- **Implementation:** Restrictions on installing unauthorized software, connecting to untrusted networks, or sharing sensitive files.
- **Example:** Netflix prohibits the use of personal cloud storage for company data.

D. Incident Response Policies:

Outline the procedures for identifying, reporting, and mitigating security incidents.

- **Implementation:** Security Information and Event Management (SIEM) systems, predefined escalation paths, and digital forensics protocols.
- **Example:** Apple has a 24/7 Computer Security Incident Response Team (CSIRT).

E. Compliance and Regulatory Policies:

Ensure adherence to global data protection laws and industry standards.

- **Implementation:** GDPR, CCPA, HIPAA compliance frameworks, regular audits, and ISO/IEC 27001 certification.
- **Example:** Facebook (Meta) implements GDPR-compliant privacy policies for EU operations.

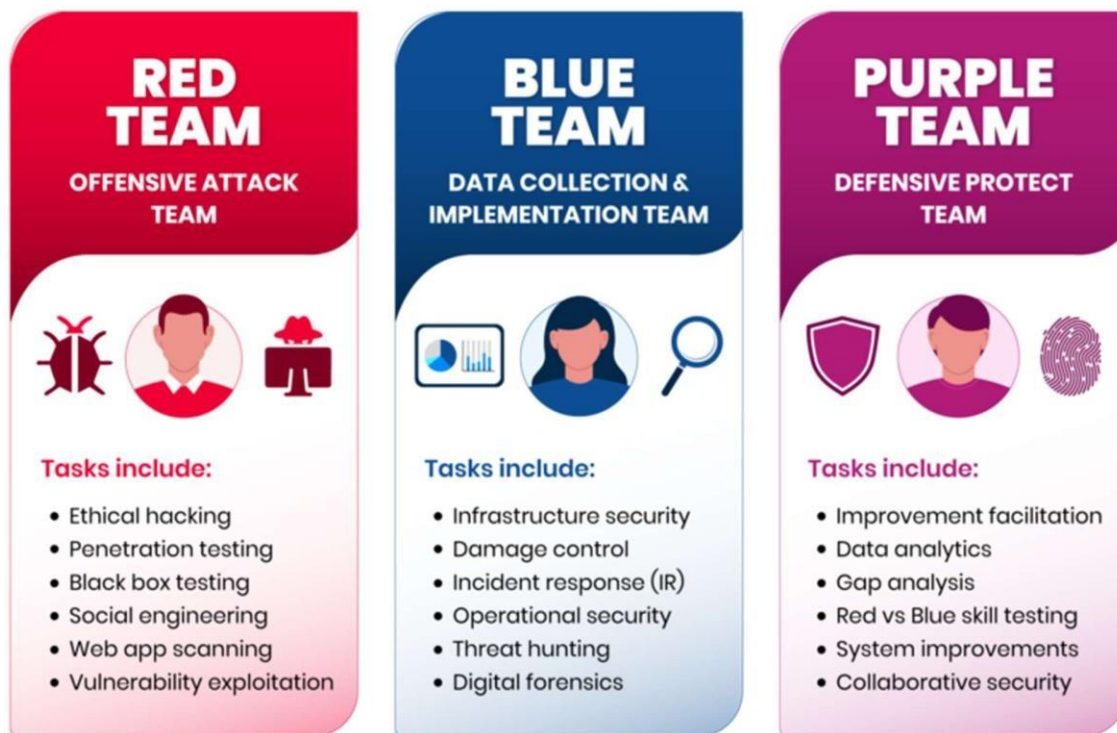
F. Remote Work and BYOD (Bring Your Own Device) Policies:

Set rules for securely accessing corporate resources from personal devices and remote locations.

- **Implementation:** VPN usage, device encryption, and endpoint monitoring.
- **Example:** Amazon Web Services requires all remote devices to pass compliance checks before connecting to internal systems.

6. Modern Cybersecurity operations (Red, Blue, Purple Teams):

Modern organizations adopt a team-based security operations model to simulate attacks, defend against them, and improve overall cyber resilience. The three primary teams — Red, Blue, and Purple — work together to strengthen security through offensive, defensive, and collaborative efforts.



A. Red Team – Offensive Security:

The Red Team acts as **ethical attackers** who simulate real-world cyber threats to identify vulnerabilities in systems, applications, and networks.

- **Objective:** Test and challenge the organization's security posture.
- **Activities:**
 - Penetration testing
 - Social engineering
 - Exploit development
 - Physical security testing

- **Example:** Conducting a simulated phishing attack to test employee awareness.

B. Blue Team – Defensive Security:

The Blue Team is responsible for defending systems and responding to attacks. They monitor for suspicious activities, analyse threats, and take actions to mitigate risks.

- **Objective:** Detect, prevent, and respond to cyberattacks in real time.
- **Activities:**
 - Threat hunting and incident response
 - Security Information and Event Management (SIEM) monitoring
 - Log analysis and forensic investigations
 - Patch management and security updates
- **Example:** Identifying and blocking a malicious IP attempting brute-force login attempts.

C. Purple Team – Collaboration & Optimization:

The Purple Team bridges the gap between Red and Blue Teams, ensuring that attack simulations and defence strategies work in harmony.

- **Objective:** Improve security by fostering knowledge sharing and process refinement.
- **Activities:**
 - Coordinating attack simulations and defensive improvements
 - Validating detection rules against Red Team activities
 - Continuous security training and drills
- **Example:** Reviewing Red Team attack findings with the Blue Team to enhance intrusion detection systems (IDS).

7. Networking in Cybersecurity:

Networking forms the foundation of cybersecurity, as all cyber threats, defences, and communications occur over interconnected systems. Understanding networking concepts is essential for securing data flow, detecting malicious activity, and mitigating attacks.



A. Role of Networking in Cybersecurity:

Networking enables devices, servers, and applications to communicate, but this connectivity also introduces risks. Cybersecurity professionals must understand network structures to:

- Identify vulnerabilities in communication channels
- Implement secure data transfer methods
- Monitor and filter traffic to prevent intrusions

B. Core Networking Components in Security:

- **IP Addressing & Subnetting:** Used for network segmentation and isolating critical systems.
- **Routing & Switching:** Secure routing prevents unauthorized redirection of data.

- **Firewalls:** Filter incoming and outgoing network traffic based on security rules.
- **Intrusion Detection/Prevention Systems (IDS/IPS):** Identify and block malicious traffic.

c. **Networking Threats in Cybersecurity:**

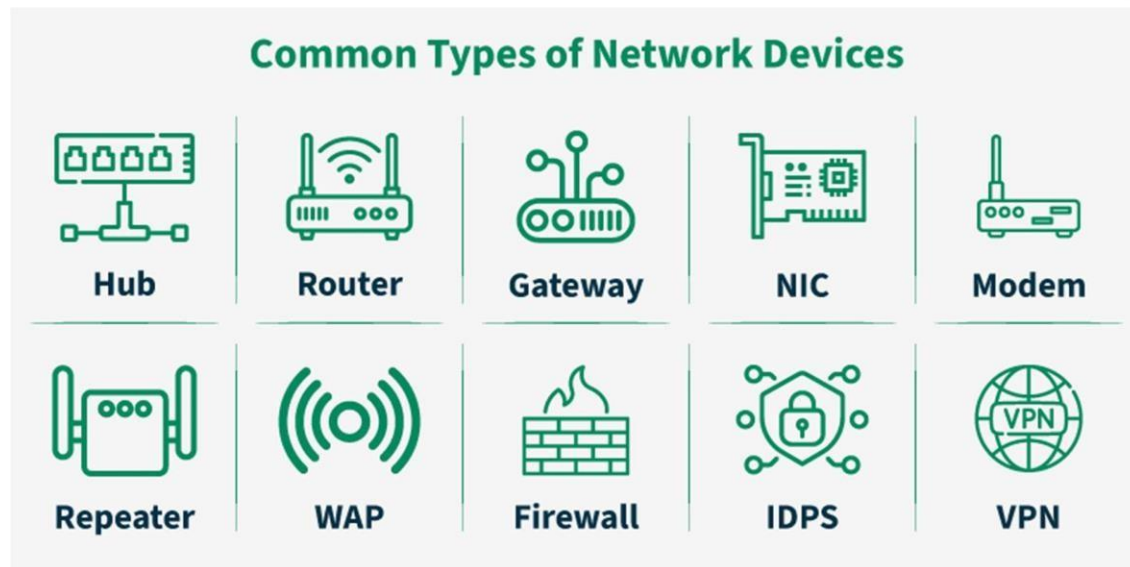
- **Man-in-the-Middle (MITM) Attacks:** Intercepting communications between two parties.
- **Packet Sniffing:** Capturing unencrypted network traffic.
- **DDoS Attacks:** Overloading network resources to cause downtime.

d. **Secure Networking Practices:**

- **Encryption:** TLS/SSL for secure web communication, VPNs for secure remote access.
- **Segmentation:** Dividing networks into smaller, isolated zones to limit breaches.
- **Access Control Lists (ACLs):** Restricting traffic based on IPs, ports, or protocols.
- **Zero Trust Networking:** Verifying every device and user before granting access.

8. **Networking Devices and Topologies:**

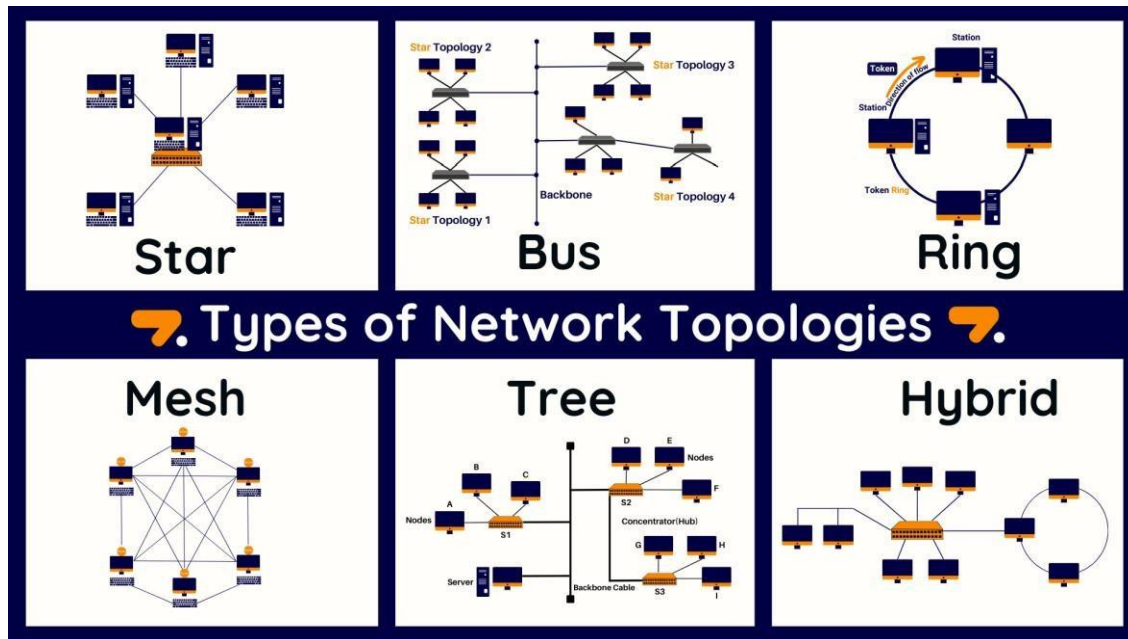
In cybersecurity, understanding networking devices and network topologies is critical for designing secure infrastructures, managing data flow, and detecting vulnerabilities. Both play a central role in how networks are structured, defended, and monitored.



A. Networking Devices:

These are hardware components that enable connectivity, data transmission, and traffic management in a network.

- **Router:** Directs data between different networks and connects LANs to the internet.
- **Switch:** Connects devices within a LAN and forwards data based on MAC addresses.
- **Hub:** Basic device that broadcasts data to all connected devices (less secure than switches).
- **Firewall (Hardware/Software):** Filters network traffic based on defined security rules.
- **Access Point (AP):** Provides wireless network connectivity to devices.
- **Gateway:** Translates data between different network protocols.
- **Load Balancer:** Distributes traffic across multiple servers for efficiency and reliability.
- **IDS/IPS Devices:** Detect and prevent suspicious network activities.



B. Network Topologies:

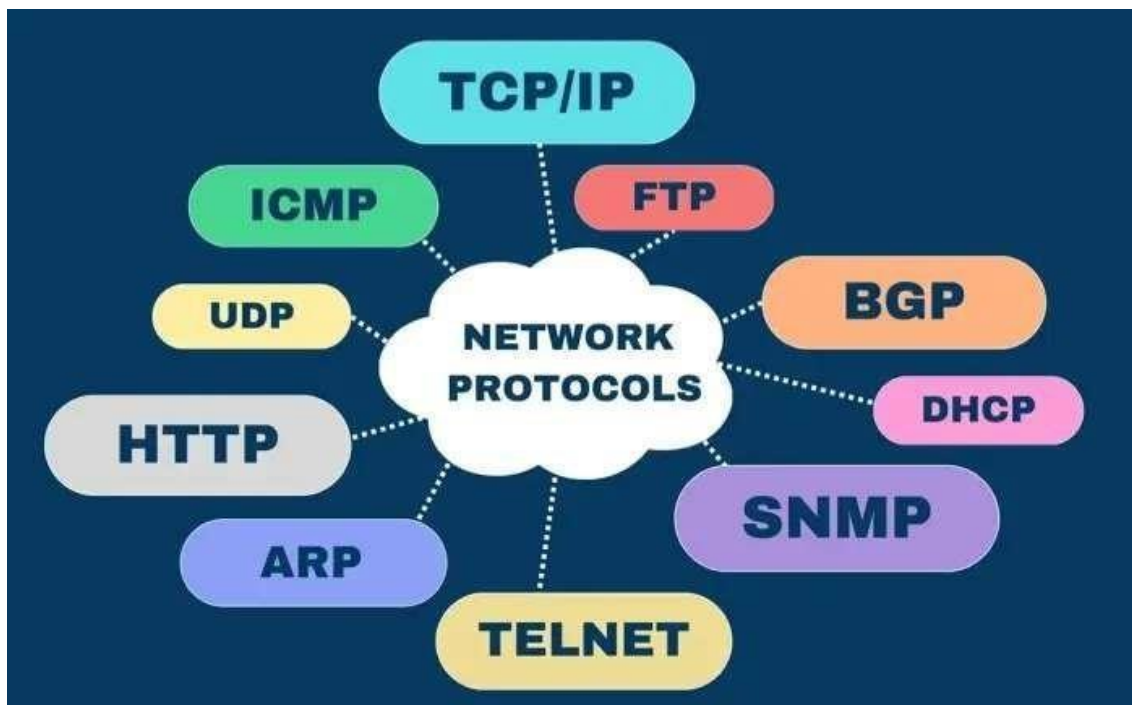
The topology defines how devices are physically or logically connected in a network.

- **Bus Topology:** All devices share a single communication line.
 - **Advantage:** Simple and inexpensive.
 - **Disadvantage:** Single point of failure.
- **Star Topology:** All devices connect to a central hub or switch.
 - **Advantage:** Easy to manage and troubleshoot.
 - **Disadvantage:** Central device failure disrupts the network.
- **Ring Topology:** Devices connect in a circular chain.
 - **Advantage:** Predictable data flow.
 - **Disadvantage:** Failure of one device can disrupt the loop.
- **Mesh Topology:** Every device connects to every other device.
 - **Advantage:** High redundancy and reliability.

- **Disadvantage:** Expensive and complex to set up.
- **Hybrid Topology:** Combines two or more topology types for flexibility and scalability.

9. Network Protocols:

Network protocols are standardized rules that define how data is transmitted, received, and interpreted across networks. They ensure interoperability between devices, security of communications, and efficiency in data exchange—making them a core component of cybersecurity.



A. Importance in Cybersecurity:

Protocols establish the language of communication between systems. Understanding them allows cybersecurity professionals to:

- Detect anomalies and malicious traffic
- Configure firewalls and intrusion detection systems
- Secure communication channels using encryption

B. Common Network Protocols

- **Communication Protocols**
 - **HTTP/HTTPS:** Transfer of web data; HTTPS encrypts traffic with SSL/TLS.
 - **FTP/SFTP:** File transfer; SFTP uses SSH for secure transmission.
 - **SMTP, POP3, IMAP:** Email communication protocols; secured with TLS for confidentiality.
- **Network Management & Security Protocols**
 - **DNS:** Resolves domain names to IP addresses; vulnerable to DNS spoofing if unsecured.
 - **DHCP:** Automatically assigns IP addresses to devices on a network.
 - **SNMP:** Manages network devices; requires encryption to prevent interception.
- **Transport & Internet Layer Protocols**
 - **TCP/IP:** Core protocols for reliable data transmission and addressing.
 - **UDP:** Faster but less reliable; often used in streaming and VoIP.
 - **ICMP:** Used for diagnostics (e.g., ping); can be exploited in DoS attacks.
- **Security Protocols**
 - **IPsec:** Encrypts and authenticates IP packets for VPNs.
 - **SSL/TLS:** Provides end-to-end encryption for secure communication.
 - **SSH:** Secure remote login and command execution.

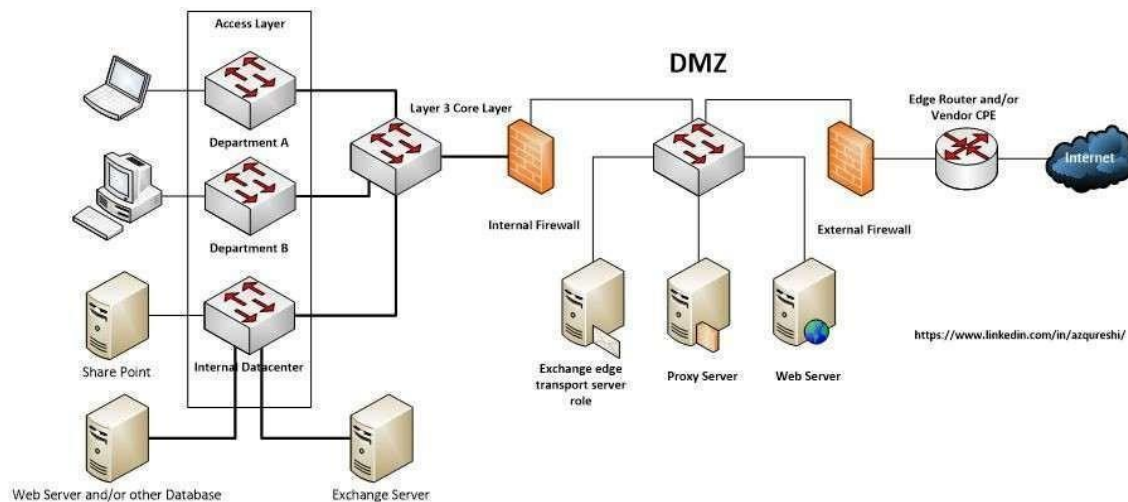
C. Protocol Vulnerabilities

- **Unencrypted Data Transmission - Risk of interception**

- **Misconfiguration** - Opens attack surfaces
- **Old Versions** - Contain unpatched exploits

10. DMZ Architecture:

A Demilitarized Zone (DMZ) is a buffer network separating an organization's internal systems from the public internet. It hosts public-facing services while limiting access to the internal LAN.



Purpose:

- Protect internal systems from direct exposure
- Allow controlled access to public services (web, email, DNS)
- Contain breaches if a public server is compromised

Key Components:

- **External Firewall:** Internet → DMZ
- **Internal Firewall:** DMZ → LAN
- **Public Servers:** Web, FTP, mail
- **Monitoring:** IDS/IPS for threat detection

Deployment Models:

- **Single Firewall:** Simpler, less secure
- **Dual Firewall:** More secure, preferred in enterprises

Advantage:

Isolates risk, enforces traffic control, and supports layered defence.

11. Information Gathering Tools:

Information gathering is the first phase of ethical hacking, focusing on collecting data about a target to identify vulnerabilities before testing. It uses passive (indirect) and active (direct) methods.



Purpose:

- Map target's network and assets
- Identify vulnerabilities
- Support penetration testing planning

Key Tools & Resources:

- **Nmap** – Network scanning & host discovery
- **WHOIS** – Domain registration details (e.g., whois.domaintools.com)
- **nslookup.io** – DNS query and record lookup
- **webchecker.xyz** – Website performance & uptime monitoring

- **Google Dorks** – Advanced search queries to find exposed data
- **exploit-db** – Public exploits & vulnerability database
- **Wappalyzer** – Detects website technologies
- **Social Media Networks** – OSINT on individuals/organizations
- **Wayback Machine** – Archived website snapshots
- **Reverse IP Lookup** – Identifies sites hosted on the same IP
- **Groups, Forums & Blogs** – Hidden intel from public discussions
- **IP Geolocation** – Physical location of an IP
- **Subdomain Finder** – Maps subdomains of a target
- **Shrewdeye.app** – OSINT platform for domain & email discovery
- **viewdns.info** – Multi-DNS and network tools
- **hackertarget.com** – Recon & vulnerability scanning tools

Security Role:

Helps uncover information that attackers could exploit, enabling proactive defence.

12. Cybersecurity File types (VDI, OVA, VMDK, ISO):

In cybersecurity, virtual machine and disk image file formats are used for labs, malware analysis, penetration testing, and safe software execution.

VDI (Virtual Disk Image):

- Format used by Oracle VirtualBox.
- Stores a virtual machine's hard disk contents.
- Ideal for portable and isolated testing environments.

OVA (Open Virtual Appliance)

- Pre-packaged VM format with all configuration files.
- Supports multiple platforms like VirtualBox & VMware.
- Commonly used to distribute prebuilt security labs (e.g., vulnerable machines).

VMDK (Virtual Machine Disk):

- VMware's virtual disk format.
- Stores complete virtual OS data.
- Often used in enterprise penetration testing setups.

4. ISO (Optical Disk Image):

- Sector-by-sector copy of a CD/DVD.
- Used for OS installations, bootable rescue disks, and security tools (e.g., Kali Linux ISO).

Security Role:

These file types enable safe, isolated environments for testing exploits, analysing malware, and practicing ethical hacking without risking real systems.

13. Network modes in Virtual Environments:

Virtualization platforms like VirtualBox and VMware provide different network modes to simulate real-world connectivity for cybersecurity testing.

	VM ↔ VM	VM → Host	VM ← Host	VM → LAN	VM ← LAN
Not attached	–	–	–	–	–
NAT	–	+	Port Forward	+	Port Forward
NAT Network	+	+	Port Forward	+	Port Forward
Bridged	+	+	+	+	+
Internal Network	+	–	–	–	–
Host-only	+	+	+	–	–

NAT (Network Address Translation)

- VM shares host's IP to access the internet.
- Secure and isolated from external access.

Bridged Mode

- VM acts as a separate device on the physical network.
- Allows direct communication with other LAN devices.

Host-Only

- VM communicates only with the host machine.

- Fully isolated from the internet.

4. Internal Network

- Communication only between VMs in the same virtual network.
- No internet or host access.

5. NAT Network

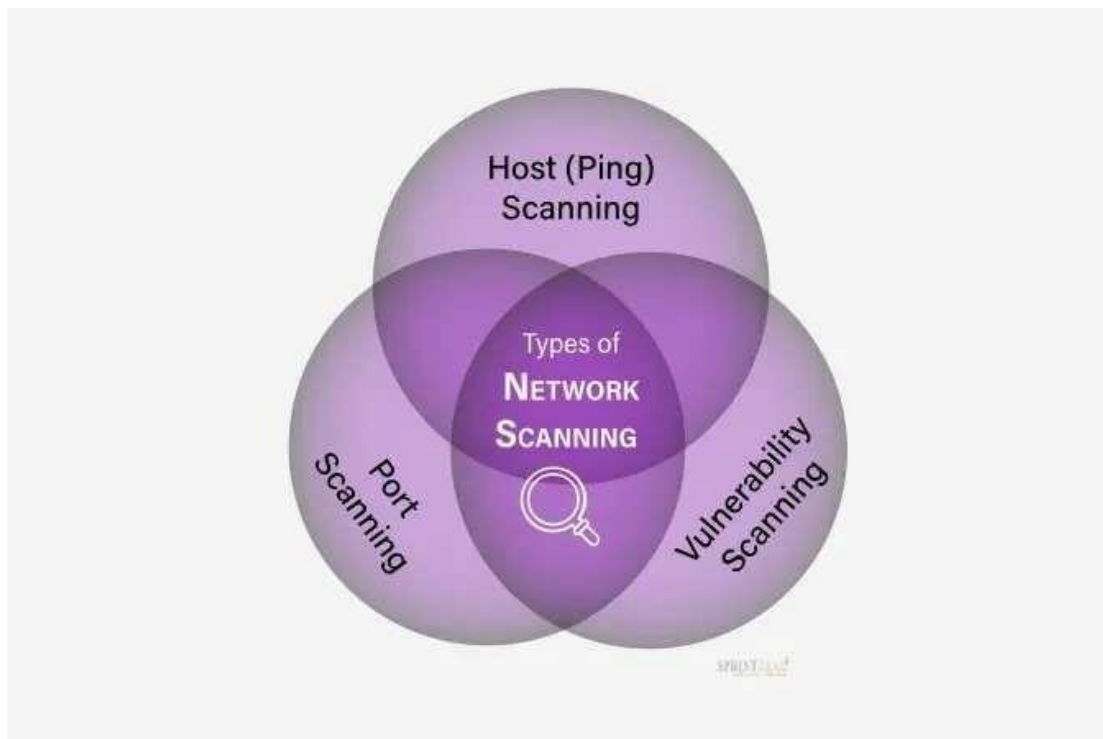
- Similar to NAT but allows multiple VMs to communicate internally.

Security Role:

Selecting the right mode ensures controlled connectivity during penetration testing, malware analysis, or training labs, balancing isolation and realism.

14. Scanning in Cybersecurity:

Scanning is the second phase of ethical hacking, following information gathering. It identifies live hosts, open ports, services, and vulnerabilities in a target network.



Types of Scanning:

- Network Scanning – Detects active devices and their IP addresses.
- Port Scanning – Identifies open, closed, or filtered ports.
- Vulnerability Scanning – Finds known security flaws in systems or applications.

Common Tools:

- Nmap – Host and service discovery.
- Nessus – Vulnerability assessment.
- OpenVAS – Open-source vulnerability scanning.

Security Role:

Scanning provides a blueprint of the target's attack surface, helping ethical hackers prioritize and plan penetration testing activities.

15. Nmap Scanning Techniques & Syntax:

Nmap (Network Mapper) is a powerful open-source tool for network discovery, port scanning, and security auditing.

Common Scanning Techniques:

- **TCP Connect Scan (-sT)** – Full TCP connection, reliable but detectable.
- **SYN Scan (-sS)** – Half-open scan, stealthier and faster.
- **UDP Scan (-sU)** – Finds open UDP ports.
- **Ping Scan (-sn)** – Checks for live hosts without full port scan.
- **Service Version Detection (-sV)** – Identifies running services and versions.
- **OS Detection (-O)** – Determines target's operating system.
- **Aggressive Scan (-A)** – Combines OS detection, versioning, and traceroute.

Basic Syntax:

nmap [options] [target]

Examples:

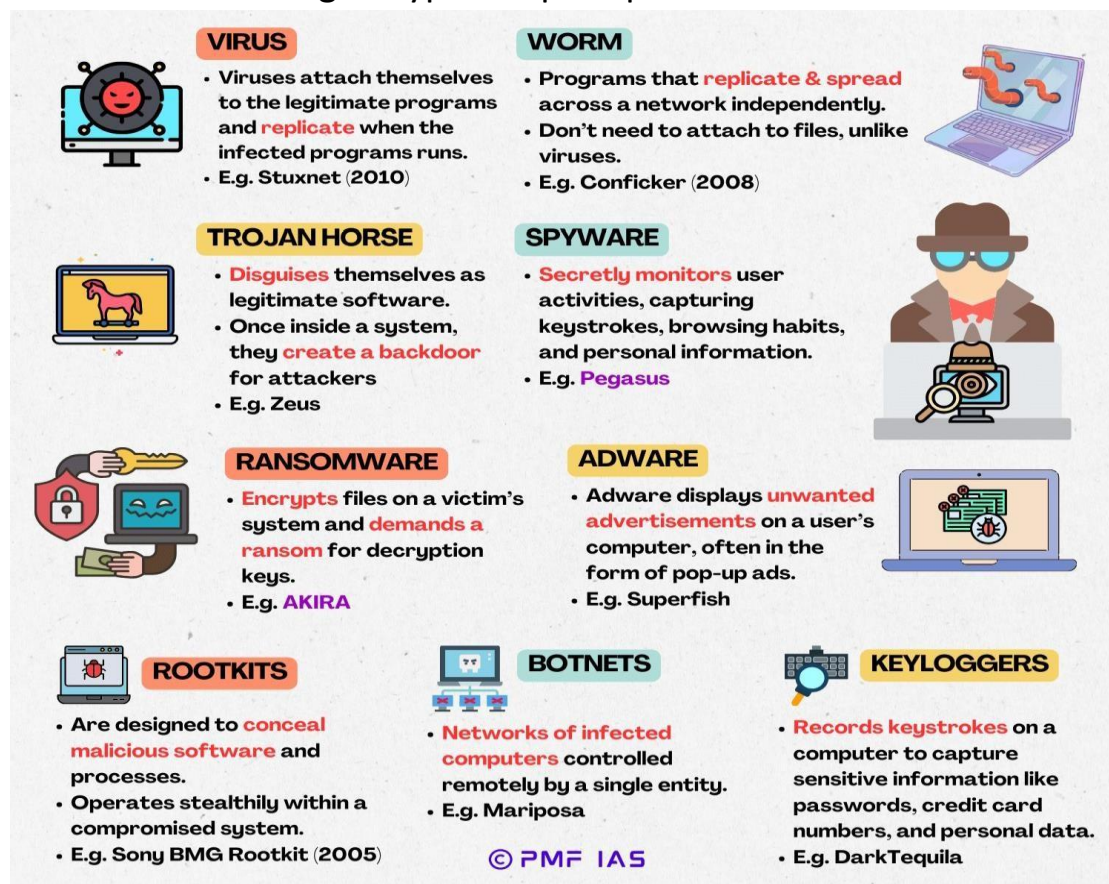
- Scan single host: nmap 192.168.1.10
- Stealth SYN scan: nmap -sS 192.168.1.10
- Service/version detection: nmap -sV 192.168.1.10
- Multiple targets: nmap 192.168.1.1-20

Security Role:

Nmap helps ethical hackers map attack surfaces and uncover vulnerable entry points before exploitation.

16. Malware Types and Definitions.

Malware (malicious software) is designed to disrupt, damage, or gain unauthorized access to systems. In cybersecurity, understanding its types helps in prevention and defence.



- **Virus** – Attaches to files/programs, spreads when infected files are executed.
- **Worm** – Self-replicating, spreads over networks without user action.
- **Trojan Horse** – Disguised as legitimate software to deliver malicious payloads.
- **Ransomware** – Encrypts data, demands payment for decryption.
- **Spyware** – Secretly monitors user activity and steals information.
- **Adware** – Displays unwanted ads, may track user behaviour.
- **Rootkit** – Hides malicious processes to maintain covert access.
- **Keylogger** – Records keystrokes to capture sensitive data.
- **Botnet** – Network of compromised devices used for coordinated attacks.

Security Role:

Identifying malware types allows targeted detection and response, reducing impact and improving system resilience.

17. DoS vs DDoS Attacks:

DoS (Denial of Service)

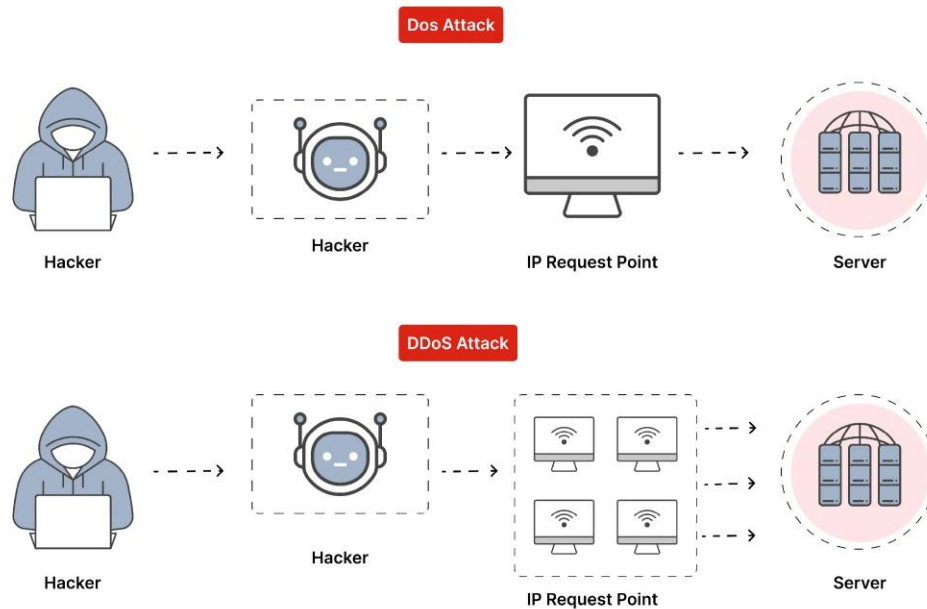
- A single source floods a target system or network with excessive requests.
- **Goal:** Overwhelm resources, making services unavailable.
- **Example:** Sending massive traffic to a web server from one machine.

DDoS (Distributed Denial of Service)

- Multiple compromised systems (botnet) coordinate to launch an attack.
- More powerful and harder to mitigate than DoS.

- **Example:** Thousands of infected devices targeting one website simultaneously.

Difference Between DoS vs DDoS



Aspect	DoS	DDoS
Source	Single attacker	Multiple attackers (botnet)
Scale	Limited	Large-scale, high traffic
Detection	Easier	Harder
Mitigation	Simpler	Complex, requires distributed defence

Security Role:

Recognizing attack type aids in deploying correct mitigation strategies like rate-limiting, traffic filtering, and anti-DDoS services.

18. Surface Web Types:

The Surface Web refers to the portion of the internet indexed by search engines and accessible without special tools or permissions.



Types:

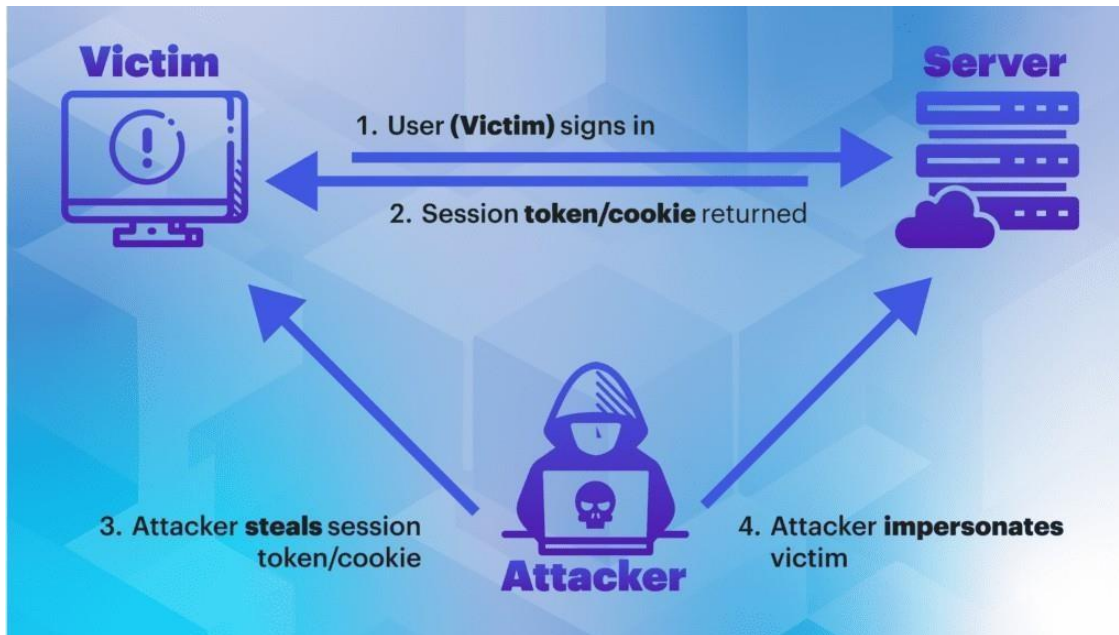
- **Static Websites** – Content remains fixed and changes rarely (e.g., company profiles, blogs).
- **Dynamic Websites** – Content generated in real-time based on user interaction (e.g., e-commerce, news sites).
- **Transactional Websites** – Enable secure transactions like banking, shopping, and booking.
- **Informational Websites** – Provide general knowledge, educational content, or public resources.

Security Role:

While publicly accessible, surface web resources can be exploited for reconnaissance, making them relevant in ethical hacking and OSINT activities.

19. Session Hijacking:

Session hijacking is a cyberattack where an attacker takes over a valid user session to gain unauthorized access to a system or service. It often exploits vulnerabilities in session management.



Types:

- **Active Hijacking** – Attacker takes control during an active session.
- **Passive Hijacking** – Attacker monitors traffic without interfering directly.

Common Methods:

- **Session Sidejacking** – Capturing session cookies over unsecured networks.
- **Cross-Site Scripting (XSS)** – Injecting malicious scripts to steal session tokens.
- **Session Fixation** – Forcing a user to use a known session ID.
- **Man-in-the-Middle (MitM)** – Intercepting and altering communication between two parties.

Security Role:

Preventive measures include HTTPS encryption, secure cookie flags, frequent session regeneration, and timeouts to limit exposure.

20. Email Spoofing Techniques:

Email spoofing is the act of forging the sender's address in an email to trick recipients into believing it's from a trusted source. It is often used in phishing attacks.



Techniques:

- **Header Manipulation** – Altering the “From” field in the email header.
- **Display Name Spoofing** – Using a legitimate name with a fake email address.
- **Domain Spoofing** – Registering domains that closely resemble a real one (typo squatting).
- **Compromised Account Usage** – Sending spoofed emails from hacked legitimate accounts.
- **SMTP Server Exploitation** – Leveraging open mail relays to send forged messages.

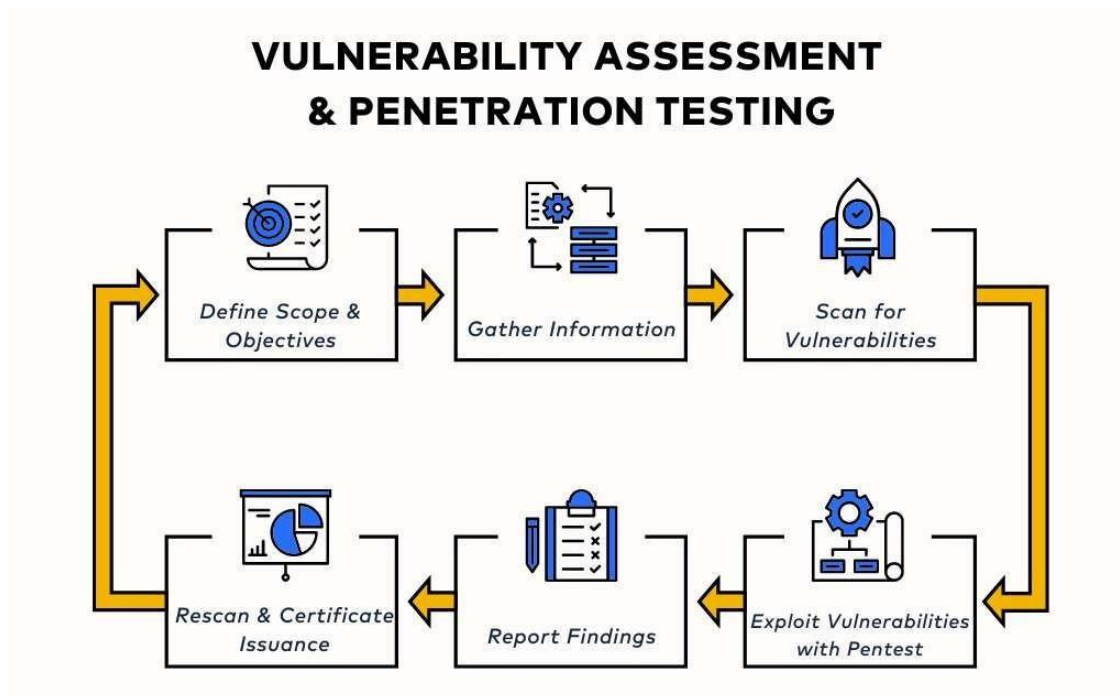
Security Role:

Mitigation includes SPF, DKIM, DMARC records, advanced

email filtering, and user awareness to identify suspicious emails.

21. VAPT (Vulnerability & Penetration Testing):

VAPT is a combined security assessment approach where Vulnerability Assessment identifies weaknesses and Penetration Testing simulates real-world attacks to exploit them.



Components:

- **Vulnerability Assessment** – Automated scanning and analysis to detect flaws.
- **Penetration Testing** – Manual and automated techniques to exploit discovered vulnerabilities.

Benefits:

- Detects security gaps before attackers do.
- Tests resilience of defences.
- Supports compliance with security standards.

Security Role:

VAPT provides a comprehensive security posture

evaluation, ensuring both detection and validation of vulnerabilities.

22. Web Application Penetration Testing:

A security assessment process focused on identifying and exploiting vulnerabilities in web-based applications to evaluate their security.



Objectives:

- Detect flaws in authentication, authorization, and input validation.
- Identify misconfigurations and insecure code.
- Assess resistance to real-world attacks.

Common Test Areas:

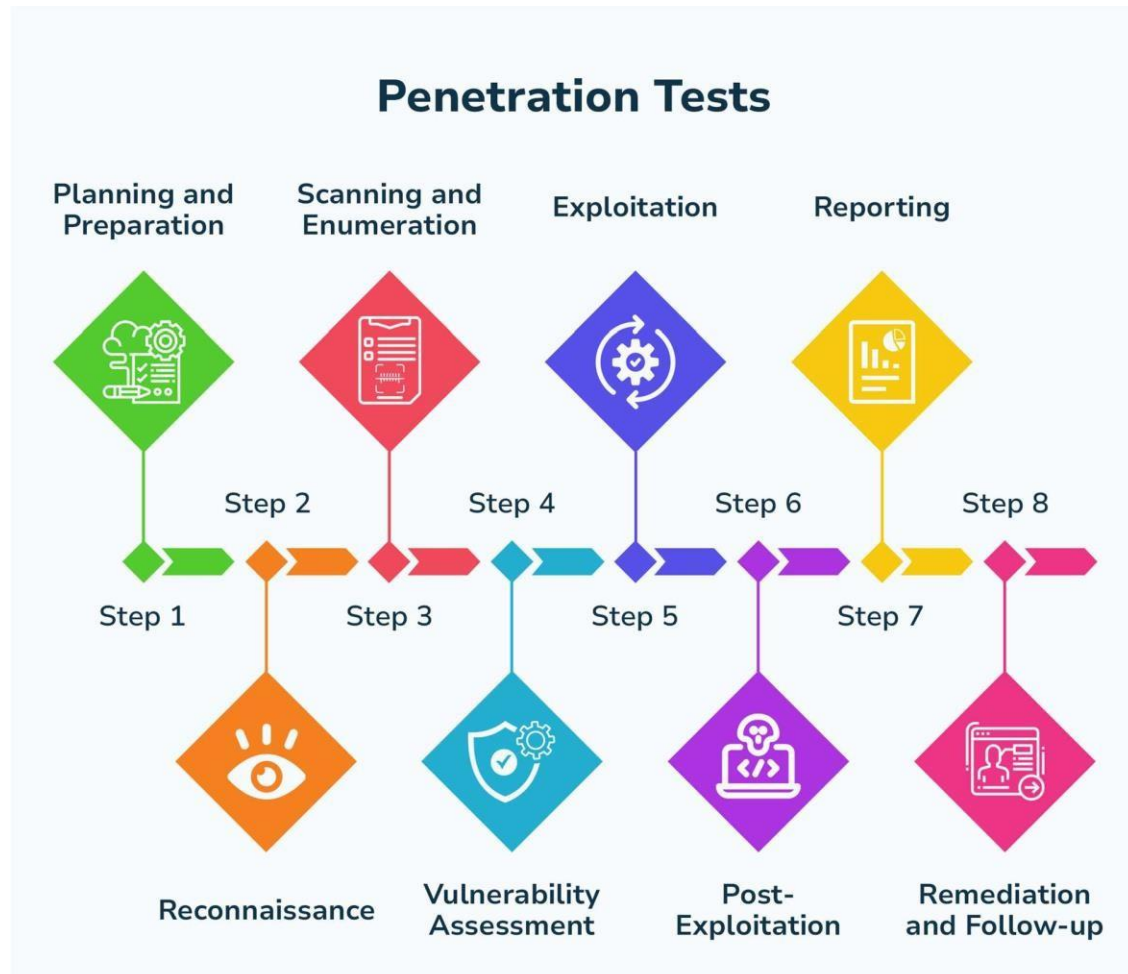
- SQL Injection (SQLi)
- Cross-Site Scripting (XSS)
- File Inclusion vulnerabilities
- Insecure Direct Object References (IDOR)
- Session Management flaws

Security Role:

Helps organizations secure web applications, protect sensitive data, and comply with security standards like OWASP Top 10.

23. Penetration Testing Methodologies & Phases:

A structured approach to simulating cyberattacks on systems, networks, or applications to identify and exploit vulnerabilities.



Common Methodologies:

- **OSSTMM** – Open Source Security Testing Methodology Manual.
- **OWASP** – Focused on web application security.
- **NIST SP 800-115** – Technical guide for security testing.
- **PTES** – Penetration Testing Execution Standard.

Phases:

- **Planning & Reconnaissance** – Define scope, gather intelligence.

- **Scanning** – Identify live hosts, services, and potential entry points.
- **Exploitation** – Attempt to breach systems using identified weaknesses.
- **Post-Exploitation** – Assess impact, maintain access if needed.
- **Reporting** – Document findings, risks, and remediation steps.

Security Role:

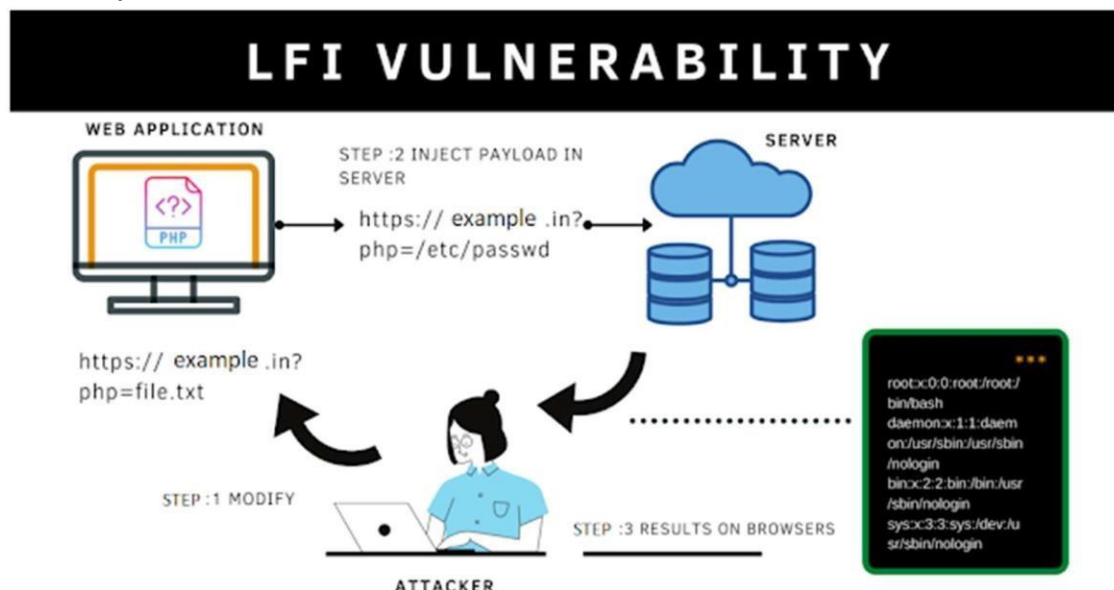
Ensures systematic, repeatable, and measurable security testing for better defence readiness.

24. File Inclusion Vulnerabilities:

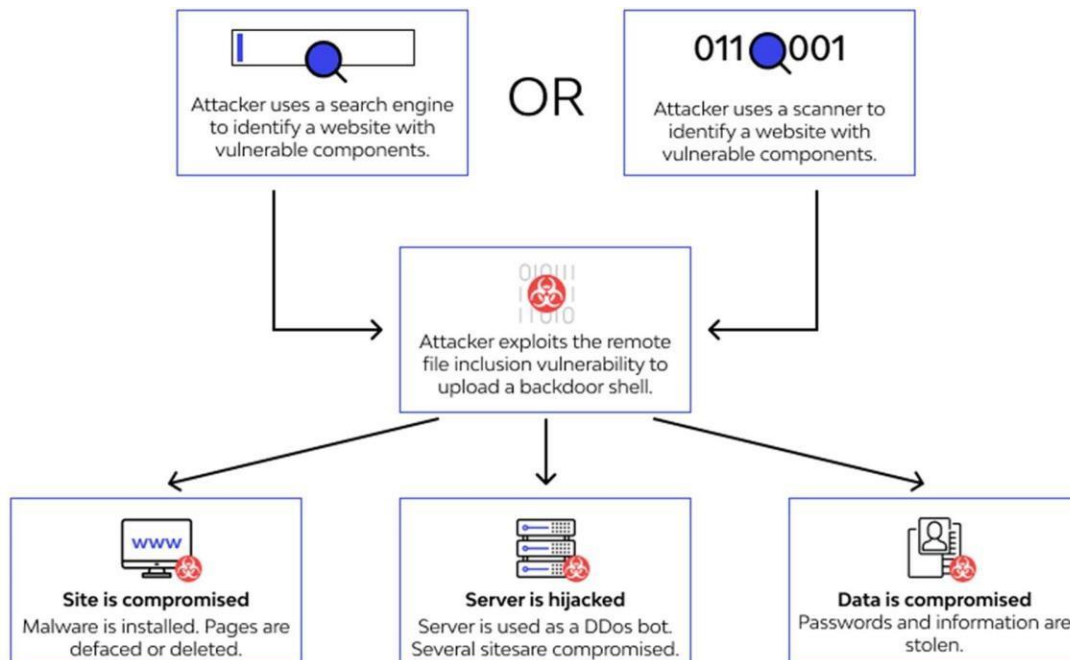
A web security flaw that occurs when applications allow unvalidated user input to control file paths, leading to the inclusion of unauthorized files.

Types:

Local File Inclusion (LFI) – Loads files from the server's local file system.



Remote File Inclusion (RFI) – Loads files from an external server over the internet.



Risks:

- Information disclosure (e.g., configuration files).
- Remote code execution.
- Privilege escalation.

Security Role:

Prevented by input validation, using whitelists, disabling risky functions, and enforcing secure coding practices.

25. Insecure Design Flaws:

Security weaknesses caused by poor system architecture or flawed security logic, rather than implementation bugs.

Examples:

- No rate-limiting on login attempts.
- Weak access control models.
- Storing sensitive data without encryption.
- Overly permissive default configurations.

Risks:

- Enables brute-force, privilege escalation, or data breaches.
- Often harder to fix because they require architectural changes.

Security Role:

Mitigated through secure design principles, threat modelling, and security by design approaches during the development lifecycle.

26. Bug Bounty Platforms:

Online platforms that connect organizations with ethical hackers to identify and report security vulnerabilities in exchange for monetary rewards or recognition.

Popular Platforms:

- **HackerOne** – Large community, government and enterprise programs.
- **Bugcrowd** – Managed bug bounty and vulnerability disclosure.
- **Synack** – Combines crowdsourced testing with AI-driven security.
- **Open Bug Bounty** – Free and open disclosure platform.

Benefits:

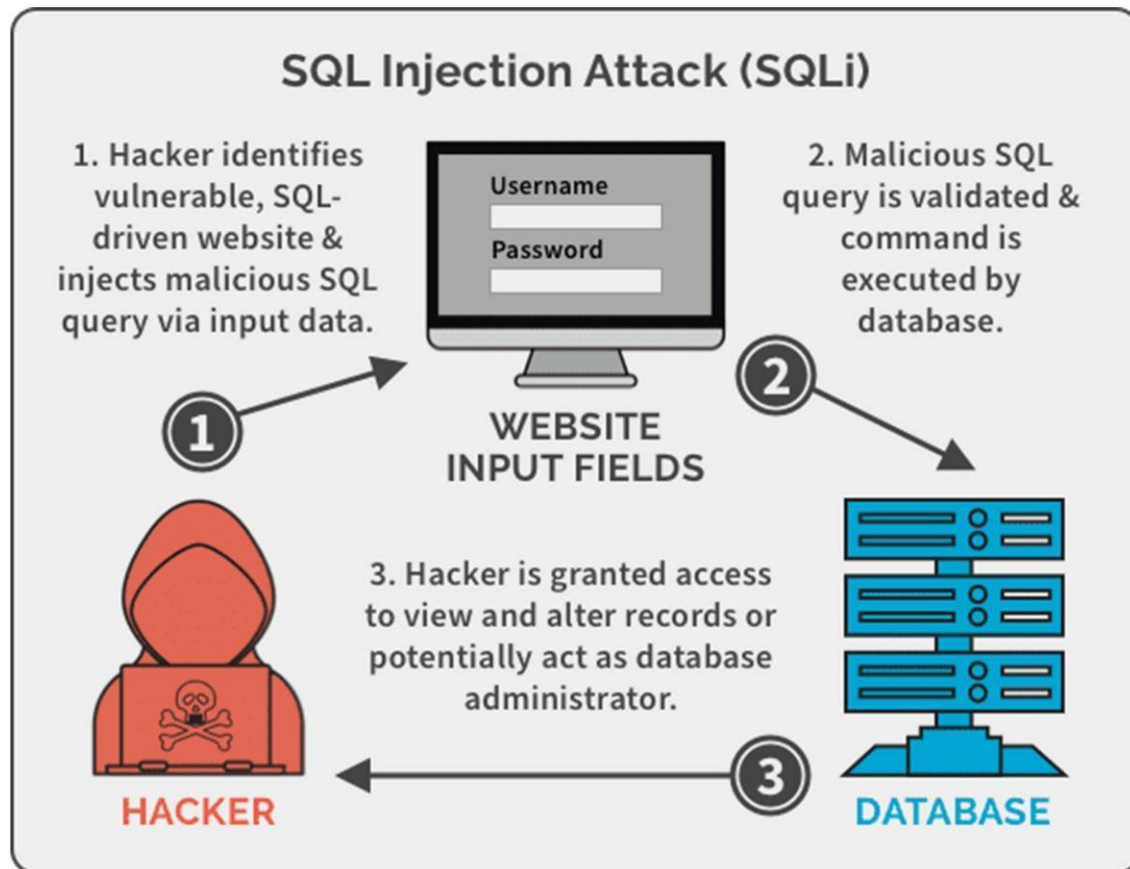
- Continuous security testing by diverse experts.
- Cost-effective compared to full-time security teams.
- Builds community trust and brand credibility.

Security Role:

Acts as a proactive security layer, leveraging global hacker expertise to find vulnerabilities before malicious actors do.

27. SQL Injection (SQLi):

A web application vulnerability where attackers inject malicious SQL queries into input fields to manipulate a database.



Types:

- **In-band SQLi** – Data retrieved using the same communication channel.
- **Blind SQLi** – No direct output; attacker infers data from responses.
- **Out-of-band SQLi** – Uses different channels to retrieve data.

Risks:

- Unauthorized data access.
- Modification or deletion of records.
- Full database compromise.

Security Role:

Prevented by using prepared statements, parameterized queries, input validation, and least-privilege database accounts.

28. Injection Attacks (HTML, iFrame ,Command Injection):

A class of attacks where untrusted input is sent to an interpreter or browser, altering intended commands or injecting malicious code.

Types:

HTML Injection – Inserting malicious HTML code into a web page to alter content or redirect users.



iFrame Injection – Embedding hidden or malicious iFrame to load harmful sites.



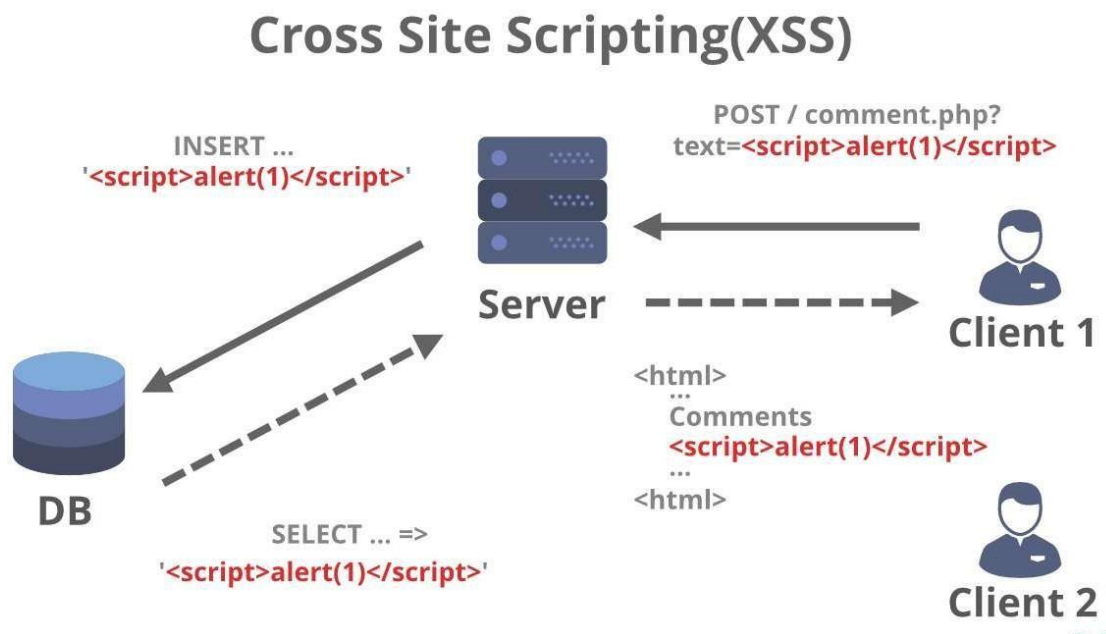
Command Injection – Executing system-level commands through vulnerable applications.

Security Role:

Prevented via input sanitization, output encoding, disabling risky functions, and strict content security policies (CSP).

29. Cross Site Scripting (XSS) Overview:

A client-side code injection vulnerability that allows attackers to run malicious scripts in users' browsers by injecting them into trusted websites.



Types:

- **Stored XSS** – Malicious script permanently stored on the server.
- **Reflected XSS** – Script delivered via a crafted URL or request.
- **DOM-based XSS** – Script injection occurs in the browser's Document Object Model without server interaction.

Risks:

- Session hijacking.

- Credential theft.
- Defacement or malicious redirection.

Security Role:

Mitigated through output encoding, input validation, CSP implementation, and secure JavaScript practices.

Conclusion:

Cybersecurity is a continuously evolving discipline that requires a deep understanding of threats, vulnerabilities, and defence strategies. From foundational concepts like the CIA triad and network protocols to advanced practices such as ethical hacking, penetration testing, and threat modelling, each component plays a critical role in securing digital assets.

Modern defence demands proactive monitoring, structured security policies, and coordinated operations between Red, Blue, and Purple Teams. Leveraging tools, frameworks, and community-driven platforms like bug bounties enhances resilience by identifying risks before adversaries exploit them.

Ultimately, cybersecurity is not a one-time effort but an ongoing process of adaptation, innovation, and vigilance to protect against ever-changing attack landscapes.